THE TWO TILDE CONSTRUCTIONS ON AN AFFINE SCHEME AGREE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1315-1327, following the statement at 1294-1313.

Editorial note. The mapping property in the statement is inherited from the named Modules engine via the constructed isomorphism, as the final sentence records.

Allowed standard prerequisites (editorial).

- Construction of quasi-coherent sheaves on ringed spaces. Modules, Lemma modules-lemma-construct-quasi-coherent-sheaves (cross-chapter, cited at ~1316). Gives the canonical map F_M -> tilde(M) and the F_M mapping property.

Source reference. Original definition 01HU: sources/stacks-schemes/pinned-original-source.tex, source lines 888–888. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BI: focused-seeds/geometry/S9-schemes-compare-constructions/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BH: focused-seeds/geometry/S9-schemes-compare-constructions/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be an affine scheme. Let M be an R-module. There exists a canonical isomorphism between the sheaf $\widetilde{M}$ associated to the R-module M (Definition 01HU) and the sheaf $\mathcal{F}_M$ associated to the R-module M (Modules, Definition 01BI). This isomorphism is functorial in M. In particular, the sheaves $\widetilde{M}$ are quasi-coherent. Moreover, they are characterized by the following mapping property*

$$\mathrm{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathcal{F}) = \mathrm{Hom}_R(M, \Gamma(X, \mathcal{F}))$$

for any sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$. Here a map $\alpha : \widetilde{M} \to \mathcal{F}$ corresponds to its effect on global sections.

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Proof. By Modules, Lemma 01BH we have a morphism $\mathcal{F}_M \to \widetilde{M}$ corresponding to the map $M \to \Gamma(X, \widetilde{M}) = M$. Let $x \in X$ correspond to the prime $\mathfrak{p} \subset R$. The induced map on stalks are the maps $\mathcal{O}_{X,x} \otimes_R M \to M_\mathfrak{p}$ which are isomorphisms because $R_\mathfrak{p} \otimes_R M = M_\mathfrak{p}$. Hence the map $\mathcal{F}_M \to \widetilde{M}$ is an isomorphism. The mapping property follows from the mapping property of the sheaves $\mathcal{F}_M$. $\square$